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LETTER

The Gravitational Constitutive Law from the Tetrad Frame Geometry

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Abstract
The AQUAL field equation requires an interpolating function $\mu(x)$ whose form has remained undetermined since Milgrom's original formulation in 1983. We derive it through three methodologically distinct routes. The first reads the graded rank structure of the metric and tetrad equation. The second derives the function from equilibrium field closure within the AQUAL framework. The third, resting on the companion letter [Nelson 2026a], expresses the exchange symmetry of the tetrad equation's two parity-related bounded regions as a functional equation on the odds ratio $\Omega = \mu/(1-\mu)$, forcing $\Omega(x) = x$ and therefore $\mu(x) = x/(x+1)$. All three routes, through different mathematics, identify the same algebraic fact: the capacity ratio between the two tetrahedral sectors equals the dimensionless acceleration. The constitutive law is the odds-to-fraction normalisation of that ratio. Its consequences include the complement symmetry $\mu(x) + \mu(1/x) = 1$, a closed-form mass decoder, and the baryonic Tully-Fisher relation as a theorem.

1 The open problem

In 1983, Milgrom [1] showed that the dynamics of galaxies at low acceleration can be captured by a single modification to the Poisson equation, provided an interpolating function $\mu(x)$ mediates between the Newtonian regime at high acceleration and a scale-invariant regime at low acceleration. Bekenstein and Milgrom [2] subsequently embedded this modification in the AQUAL field equation,

$$\nabla \cdot \left[\mu\left(\frac{|\nabla\Phi|}{a_0} \right) \nabla\Phi \right] = 4\pi G\rho, \tag{1}$$

which derives from an action principle and therefore conserves energy, momentum, and angular momentum. The function $\mu(x)$, where $x = |\nabla\Phi|/a_0$, must approach unity at large x to recover Newtonian gravity and must scale linearly with x as $x \to 0$ to produce flat rotation curves.

Infinitely many functions satisfy both limits. Among the candidates explored in the literature, the "simple" form $x/(x+1)$ [3], the "standard" form $x/\sqrt{1+x^2}$ [1], and transcendental forms such as $1 - e^{-\sqrt{x}}$ all produce acceptable rotation curves. Famaey and Binney [3] showed that the simple form provides the best fit to the Milky Way terminal velocity curve, but selection by fitting is not derivation. A derivation of μ from first principles has remained open for four decades.

The present letter derives it. The derivation rests on the coordinate-space geometry of the tetrad frame established in the companion letter [4], which proved that the tetrad equation $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ produces two parity-related bounded tetrahedral regions whose union is a stellated octahedron. The exchange symmetry between those two regions provides one of three routes to the constitutive law developed here.

2 The law

The gravitational constitutive law is

$$\boxed{\mu(x) = \frac{x}{x+1}.} \tag{2}$$

It is the unique function consistent with the exchange symmetry of the two parity-related sectors of the tetrad equation, under rational bijectivity on the projective line, as proved in Section 5 and the accompanying Supplementary Note 3.

Three distinct derivational routes, beginning from different starting points and using different operative mathematics, produce this function. All three ultimately identify the same ratio law, $\mu/(1-\mu) = x$, from which $\mu = x/(x+1)$ follows by normalisation. The following three sections present them in turn.

3 From the metric

The internal metric of the tetrad formalism, $\eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$, contains two numbers readable by inspection: the frame dimension $K = 4$ (the matrix is 4×4) and the spatial dimension $D = 3$ (three positive eigenvalues). Because the tetrad equation is degree 2 in e, a third number follows: the root count $n = 2$, whose geometric realisation as two parity-related bounded regions is established in the companion letter [4].

These three numbers satisfy a closure relation. Two simplices with $D+1$ faces each contribute $2(D+1)$ faces; D binary spatial axes contribute 2^D sign combinations. At $D = 3$ the

two counts are equal:

$$2(D+1) = 2^D = 8. \tag{3}$$

This equality holds at $D = 3$ as the minimum threshold and admits no other positive integer solution, since 2^D grows exponentially while $2(D+1)$ grows linearly.

From the same two integers, the tetrad equation's graded structure can be read. A scalar closure condition $\det(e) \neq 0$ carries rank 0 on the frame space (dimension $K^0 = 1$). The frame $e^a{}_\mu$ carries rank 1 (dimension $K^1 = 4$). The bilinear coupling $\eta_{ab}\, e^a\, e^b$ carries rank 2 (dimension $K^2 = 16$). Taking the graded dimension of a rank-r ingredient to be K^r, the natural three-term encoding is the coupling polynomial:

$$f(K) = 1 + 4 + 16 = 21. \tag{4}$$

Under spatial parity, a rank-r tensor transforms with sign $(-1)^r$ [5], so ranks 0 and 2 are even while rank 1 is odd. Because the stellated octahedron has exactly three structural components (TetA, the octahedral core, and TetB), these define a natural three-state parity coordinate $s \in \{+1, 0, -1\}$, with TetA and TetB as the parity-opposite extrema and the octahedral core as the symmetric midpoint. The parity-odd contribution enters only the rank-1 term, and interpolation on three points with odd symmetry fixes this contribution uniquely as Ks:

$$f(K, s) = 1 + sK + K^2. \tag{5}$$

The polynomial evaluates to $f = 21$ at TetA ($s = +1$), $f = 17$ at the octahedral core ($s = 0$), and $f = 13$ at TetB ($s = -1$), with range $\Delta f = 2K = 8$.

To make the discrete state parameter continuous in the acceleration ratio $x = g/a_0$, a map $s(x)$ is needed. The map must be rational, to preserve the polynomial's algebraic structure, and bijective, so that each acceleration corresponds to exactly one geometric state. A rational bijection on the projective line is necessarily fractional linear [6]. Three boundary conditions ($s \to +1$ as $x \to \infty$; $s = 0$ at $x = 1$; $s \to -1$ as $x \to 0$) determine the unique map $s(x) = (x-1)/(x+1)$. Composing with the polynomial and normalising by the range $\Delta f = 8$ yields $\mu(x) = x/(x+1)$.

The full derivation, including the Diophantine verification table and the consistency check that K and D recover the principal dimensional and combinatorial properties of $g_{\mu\nu}$, appears in Supplementary Note 1. This route uses no field equation and no equilibrium postulates; it reads the constitutive law from the algebraic content of the metric and the tetrad equation.

4 From equilibrium

The second derivation begins from the AQUAL field equation itself, which provides the Lagrangian context for modified dynamics. If the gravitational field is modelled as a single-channel processor with finite capacity, the simplest closure consistent with the asymptotic limits is the minimal single-channel saturating closure (MSC) class. Within MSC, three physical postulates constrain the field's response: Newtonian correspondence at high acceleration, finite encoding capacity producing scale invariance at low acceleration, and single-channel saturation governed by detailed balance.

At each point, two opposing propensities govern the partition between a resolved fraction ϵ tracking Newtonian demand and an unresolved fraction $1 - \epsilon$ at saturation. The resolution propensity scales as $x^n(1-\epsilon)$ and the relaxation propensity scales as ϵ, where n is the driving exponent. At equilibrium, these balance to give $\epsilon = Kx^n/(1 + Kx^n)$, with $K = \kappa_+/\kappa_-$ the rate ratio. In the small-x limit, $\mu \approx Kx^n$, so the deep-field scaling $\mu/x \to$ const requires $n = 1$. This is the same exponent that produces flat rotation curves and the baryonic Tully-Fisher relation $v^4 = GMa_0$.

With $n = 1$ established, the equilibrium occupancy becomes $\epsilon = Kx/(1 + Kx)$. The rate ratio K appears only in the combination Kx and can be absorbed by redefining $a_0 \to a_0/K$; this is a reparametrisation of the acceleration scale, not a new degree of freedom. The derivation therefore fixes the functional form of μ while leaving the absolute value of the crossover scale to observation. In rescaled units,

$$\mu(x) = \frac{x}{x+1}. \tag{6}$$

The equilibrium is globally attracting, with strictly negative slope in the net rate as a function of ϵ.

The full derivation, including stability analysis and the corroborating rotation-curve check, appears in Supplementary Note 2. This route does not require the coupling polynomial, rank counting, or the exchange-symmetry derivation; it reads the constitutive law from the field's equilibrium response under capacity saturation.

5 From exchange symmetry

The companion letter [4] proved that the spatial sector of the tetrad equation produces exactly two bounded tetrahedral regions, TetA and TetB, related by spatial parity. Let $\mu(x)$ denote the fraction of total field capacity carried by TetA at dimensionless acceleration $x = g/a_0$, so that $1 - \mu(x)$ is the fraction carried by TetB.

Spatial parity exchanges the two sectors, inverting the acceleration ratio and swapping the fractions:

$$\mu(1/x) = 1 - \mu(x). \tag{7}$$

The natural variable for a two-sector partition is the odds ratio $\Omega = \mu/(1-\mu)$, which measures how many times more capacity TetA carries than TetB. Under the exchange symmetry, this ratio inverts when the acceleration inverts:

$$\Omega(1/x) = \frac{1}{\Omega(x)}. \tag{8}$$

The endpoint conditions are $\Omega(0) = 0$ (the TetB limit as $x \to 0$) and $\Omega \to \infty$ as $x \to \infty$ (the TetA limit at large acceleration).

The map $\Omega(x)$ must be rational, to preserve the algebraic structure of the tetrad equation, and bijective, so that a single acceleration decodes to exactly one capacity ratio. A rational bijection on the projective line is necessarily fractional linear [6], giving the general form $\Omega(x) = (ax+b)/(cx+d)$. The endpoint condition $\Omega(0) = 0$ requires $b = 0$, reducing the map to $\Omega(x) = ax/(cx+d)$. Substituting into the exchange condition $\Omega(1/x) = 1/\Omega(x)$ and cross-multiplying gives $a^2 x = (c+dx)(cx+d)$. Comparing coefficients of x^2, x, and x^0 yields $cd = 0$ and $a^2 = c^2 + d^2$. If $d = 0$, then $\Omega = a/c$ is a constant, which is not bijective. If $c = 0$, then $a^2 = d^2$, so $a = \pm d$. The branch $a = -d$ gives $\Omega(x) = -x$, which is incompatible with $\Omega > 0$ for $x > 0$; therefore $a = d$ and

$$\Omega(x) = x. \tag{9}$$

The odds ratio between TetA and TetB is the acceleration ratio itself. From this, the constitutive law follows by normalisation:

$$\mu(x) = \frac{\Omega}{1+\Omega} = \frac{x}{x+1}. \tag{10}$$

Two properties often treated as independent conditions are in fact derived consequences. The midpoint value $\mu(1) = 1/2$ follows directly: at the acceleration scale a_0, the two sectors contribute equally, as the exchange symmetry requires. The complement symmetry $\mu(x) + \mu(1/x) = 1$ is the duality condition (7) itself, which was the starting point of the derivation rather than an additional assumption.

The full derivation, including the explicit coefficient comparison and the screening identity, appears in Supplementary Note 3.

6 Why they agree

Three derivations using different mathematics arrive at the same function. That convergence has a simple explanation: each route, through its own formalism, discovers the same algebraic fact.

In Route 1, the polynomial normalisation $(f-13)/(21-f)$ evaluates to x when $f = (21x+13)/(x+1)$. In Route 2, the equilibrium condition $\epsilon/(1-\epsilon) = x$ is the definition of the odds ratio. In Route 3, the exchange symmetry directly produces $\Omega(x) = x$. All three identify the capacity ratio between TetA and TetB with the dimensionless acceleration. The constitutive law is the odds-to-fraction normalisation of that ratio (the logistic map $\mu = \Omega/(1+\Omega)$):

$$\frac{\mu}{1-\mu} = x \qquad \Longrightarrow \qquad \mu = \frac{x}{x+1}. \tag{11}$$

The routes are methodologically distinct: Route 1 uses the coupling polynomial, Route 2 uses the AQUAL field equation, and Route 3 uses the odds ratio. No two share the same primary derivational mechanism. Their convergence reflects the fact that a single physical quantity, the TetA-to-TetB capacity ratio, is accessible through the algebraic structure of the metric, through the field's equilibrium kinetics, and through the geometry's exchange symmetry.

7 Consequences

The constitutive law carries several consequences that have previously been observed empirically but can now be traced to the TetA/TetB capacity structure derived in the three routes.

Complement symmetry. The relation $\mu(x) + \mu(1/x) = 1$ states that whatever fraction of field capacity TetA carries at acceleration x, TetB carries at $1/x$. This property was noted by Milgrom [1] as a structurally significant feature of the simple interpolating function. In the present derivation it is not a property the function happens to possess; it is the exchange symmetry of the two tetrahedral sectors, which was the starting point of Route 3.

Exact mass decoder. Because $\mu(x) = x/(x+1)$ is fractional linear, it admits an exact algebraic inverse: $x = \mu/(1-\mu)$. In spherical symmetry, substituting into the AQUAL field equation yields a closed-form expression for the baryonic mass enclosed within radius r:

$$M(r) = \frac{r\,v^4}{G\,(a_0\,r + v^2)}, \tag{12}$$

where $v(r)$ is the observed circular velocity. This decoder maps observed kinematics directly to baryonic mass without dark matter and without numerical root-finding. Transcendental

or higher-degree rational alternatives to μ generally forfeit this exact invertibility.

Baryonic Tully-Fisher relation. In the deep-field regime ($x \ll 1$), the constitutive law reduces to $\mu \approx x$, which corresponds to the TetB-dominated sector. In spherical symmetry, this limit gives $v^4 = GMa_0$: the baryonic Tully-Fisher relation [7]. One of the strongest empirical regularities in extragalactic astronomy follows from the deep-field limit of the constitutive law, which is itself the TetB sector of the tetrad geometry.

8 Discussion

The present letter derives the functional form of μ. It does not derive the value of a_0. The acceleration scale enters as an empirical input that defines the boundary between the TetA-dominated and TetB-dominated regimes; determining it from the framework's structural constants is the subject of ongoing work.

The integers {21, 17, 13} that appear in Route 1 as the three evaluation states of the coupling polynomial are developed in the companion geometry letter [4]. The constitutive law itself requires none of them: Route 3 derives it from exchange symmetry alone, and Route 2 derives it from equilibrium kinetics with no reference to the tetrad geometry. Route 1 contributes what the other two cannot: it connects the constitutive law to the metric's dimensional structure and explains why $K = 4$ and $D = 3$ appear in the polynomial. Each route carries a distinct explanatory role.

The constitutive law $\mu(x) = x/(x + 1)$ applies wherever the AQUAL field equation applies: to any gravitating system in which the acceleration passes through the scale a_0. Its primary observational arena is galaxy rotation curves, where it was first identified empirically by Milgrom [1] and confirmed by Famaey and Binney [3] through fitting to the Milky Way terminal velocity curve.

The function that has been empirical for forty years now has a derivation. It is the odds-to-fraction normalisation of the TetA-to-TetB capacity ratio, recovered through exchange symmetry, equilibrium closure, and the graded structure of the metric.

References

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